Developmental & epileptic encephalopathies Indian data: a genetic cause in 1 in 3, and a report that helps choose the medicine

For sales and healthcare teams. Facts summarised in our own words from published sources (links below). Treatment decisions are always the doctor's.

Indian study: *Genotype-phenotype correlates of infantile-onset developmental & epileptic encephalopathy syndromes in South India: a single centre experience.* Epilepsy Research 2020;166:106398 (PMID 32593896). Comprehensive epilepsy care referral centre, Thiruvananthapuram, Kerala. Author manuscript open on UCL Discovery.

Who was tested

- 82 children with DEE and seizure onset at or before 12 months (2016-2018)
- Next-generation sequencing in 69, MLPA in 13
- Unexplained refractory seizures with developmental delay

What was found

- **31.7% (26/82)** pathogenic or likely pathogenic; 84.6% of these began by 6 months
- Dravet/Dravet-like 36.4%; migrating focal seizures of infancy 67% (2/3); West syndrome 13.3%
- Commonest gene: **SCN1A**; also STXBP1, SCN2A, SCN8A, KCNT1, KCNQ2, CDKL5, ALDH7A1, GLDC and others

How the result changed treatment (as reported)

SCN1A: lamotrigine and carbamazepine avoided. **SCN2A, SCN8A:** high-dose sodium channel blockers helped infants. **Pyridoxine-dependent epilepsy:** vitamin B6 normalised development in two children. **Glycine encephalopathy, D2-hydroxyglutaric aciduria:** diet and cofactors improved seizure control. In all, **18 children could be offered targeted therapy**. The authors recommend genetic sequencing in DEE when MRI and metabolic tests are normal, with careful phenotyping.

Definitions and guideline

- **DEE (ILAE 2022):** an epilepsy with developmental impairment due to the underlying cause, the epileptic activity, or both. Examples: Dravet syndrome, Lennox-Gastaut syndrome, early-infantile DEE.
- **Testing (NSGC guideline):** genetic testing for every unexplained epilepsy; first-tier exome or genome, or an epilepsy panel of more than 25 genes; then chromosomal microarray.
- Prenatal testing in a later pregnancy: disease only; fetal sex never disclosed (PCPNDT).

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DEE: how to approach doctors

Topic	What to do or say
Doctors to visit	Paediatric neurologists, epilepsy centres, paediatricians, child development and therapy centres.
Which children	Long seizures with fever in the first year; seizures not controlled by medicines; delayed development or loss of skills; normal MRI and metabolic tests.
Opening question	"Doctor, in Dravet-like seizures, do you test SCN1A before choosing medicines?"
Facts to quote	Kerala: cause found in 31.7% of 82 children; 36.4% in Dravet-like seizures; about 1 in 5 (18 of 82) could be offered targeted therapy.
Test to suggest	Exome or genome, or an epilepsy panel of more than 25 genes, first; microarray next. Trio if possible.
"The diagnosis is clinical"	"Yes; the genetic report confirms it and helps choose medicines."
"Too expensive"	"One exome or panel can replace many separate tests." Confirm price and report time first.
Sample	EDTA blood; parents' samples for trio; counselling and written consent first.
Never say	That the test will control the seizures. Medicine changes are always the doctor's decision.

References

- Genotype-phenotype correlates of infantile-onset developmental & epileptic encephalopathy syndromes in South India: a single centre experience. Epilepsy Res 2020;166:106398. discovery.ucl.ac.uk/id/eprint/10111246
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- Srivastava P, Bhardwaj C, Mandal K. Developmental and epileptic encephalopathies: need for bridging the gaps between clinical syndromes and underlying genetic etiologies. Indian J Pediatr 2025;92:52-60.

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For general awareness of healthcare and sales teams. Not a substitute for clinical judgement; facts and guidelines may change.